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(%i1) load("C:/Users/Shruti/Desktop/Niket Aggarwal/bsc hons cs/mbe5.mac");
(%o1) "C:/Users/Shruti/Desktop/Niket Aggarwal/bsc hons cs/mbe5.mac"

(%i3) eq1:[2·x+3·y=8,x-y=1];
(%o3) [3*y+2*x = 8,x-y = 1]

(%i4) var2:[x,y];
(%o4) [x,y]

(%i6) const1:[8,1];
(%o6) [8,1]

(%i7) coef1:coefmatrix(eq1,var2);
(%o7) matrix([2,3],[1,-1])

(%i8) gauss_jordan(coef1,const1,var2);
(%o8) [x = 11/5,y = 6/5]

(%i9) eq2:[3·x-2·y=7,x+y=4];
(%o9) [3*x-2*y = 7,y+x = 4]

(%i10) const2:[7,4];
(%o10) [7,4]

(%i11) coef2:coefmatrix(eq2,var2);
(%o11) matrix([3,-2],[1,1])

(%i12) gauss_jordan(coef2,const2,var2);
(%o12) [x = 3,y = 1]

(%i13) eq3:[x+y+z=6,2·x-y+z=3,x+2·y-z=3];
(%o13) [z+y+x = 6,z-y+2*x = 3,-z+2*y+x = 3]

(%i14) var3:[x,y,z];
(%o14) [x,y,z]

(%i15) const3:[6,3,3];
(%o15) [6,3,3]

(%i16) coef3:coefmatrix(eq3,var3);
(%o16) matrix([1,1,1],[2,-1,1],[1,2,-1])

(%i17) gauss_jordan(coef3,const3,var3);
(%o17) [x = 9/7,y = 15/7,z = 18/7]

(%i19) eq4:[2·x+y-z=8,-x+3·y+2·z=3,3·x+2·y+z=13];
(%o19) [-z+y+2*x = 8,2*z+3*y-x = 3,z+2*y+3*x = 13]

(%i20) const4:[8,3,13];
(%o20) [8,3,13]

(%i22) coef4:coefmatrix(eq4,var3);
(%o22) matrix([2,1,-1],[-1,3,2],[3,2,1])

(%i25) gauss_jordan(coef4,const4,var3);
(%o25) [x = 3,y = 2,z = 0]

(%i26) eq5:[x+2·y+3·z=14,2·x-y+z=5,3·x+y+2·z=10];
(%o26) [3*z+2*y+x = 14,z-y+2*x = 5,2*z+y+3*x = 10]

(%i27) const5:[14,5,10];
(%o27) [14,5,10]

(%i29) coef5:coefmatrix(eq5,var3);

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(%i30) gauss_jordan(coef5,const5,var3);
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(%o30) [x = 3/10,y = 1/10,z = 9/2]
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